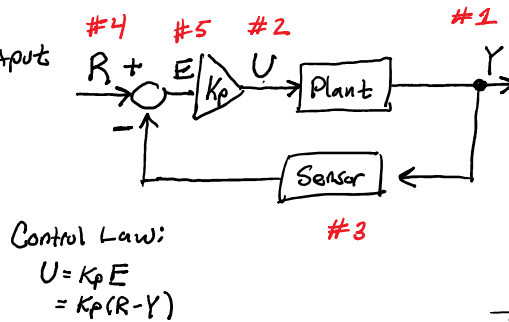


Block diagram $\leftrightarrow$ equation

How to draw a Block Diagram for a Control System

- 1) What are you trying to control
- 2) How are you going to change this output
- 3) draw the feedback loop (sensor)
- 4) Compare to reference
- 5) error & controller

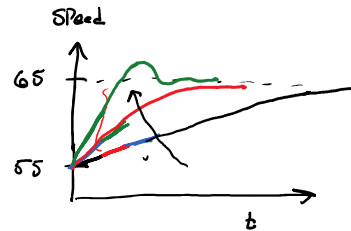
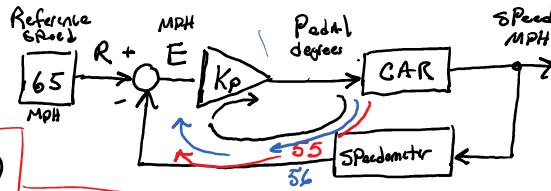

 $\vdash \rightarrow \nabla$

Controlling the Speed of a Car

Control Law:

$$\text{Pedal} = K_p E \quad (\text{generic})$$

$$\text{Pedal} = K_p (\text{Ref. speed} - \text{speed}) \quad (\text{specific})$$



Design #1

$$K_p = 0.1 \frac{\text{deg}}{\text{mph}}$$

$$\text{Pedal} = K_p E = K_p (\text{Ref. speed} - \text{speed})$$

$$= 0.1 \frac{\text{deg}}{\text{mph}} (65 \text{ mph} - 55 \text{ mph}) = 1 \text{ deg.}$$

$$\text{Pedal} = (0.1) \frac{\text{deg}}{\text{mph}} (65 - 56) = (0.1) (9 \text{ mph}) = 0.9 \text{ deg.}$$

$$\text{Pedal} = (0.1) (8.9 \text{ mph}) = 0.89 \text{ deg.}$$

⋮

⋮

$\uparrow K_p \quad \uparrow \text{speed}$
 $\downarrow \text{stability}$

Design #2

$$K_p = 1 \frac{\text{deg}}{\text{mph}}$$

$$\text{Pedal} = K_p E = (1)(10) = 10 \text{ deg.}$$

$$= (1)(7) = 7 \text{ deg.}$$

⋮

⋮

Design #3

$$K_p = 6 \frac{\text{deg}}{\text{mph}}$$

$$\text{Pedal} = K_p E = (6)(10) = 60 \text{ deg.}$$

⋮

⋮

 $K_{p\text{max}} = ?$

Design #4

$$K_p = 20 \frac{\text{deg}}{\text{mph}}$$

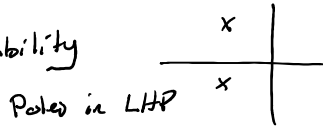
$$\text{Pedal} = K_p E = (20 \frac{\text{deg}}{\text{mph}})(10 \text{ mph}) = 200 \text{ deg.}$$

* Saturation *
 (NonLinear)

Question 1, 2, 3

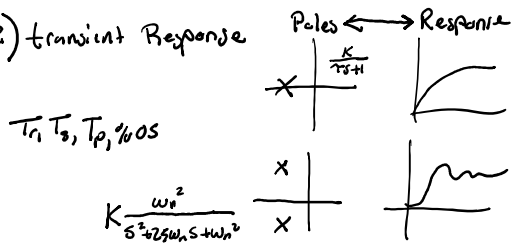
Performance of a Control System:

1) Stability

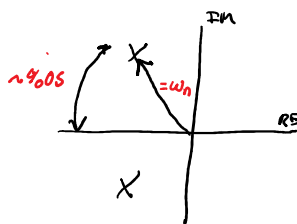


G, H

2) transient Response

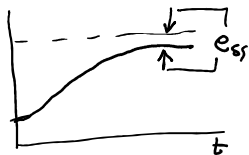


Pole Location $\leftrightarrow$ Response

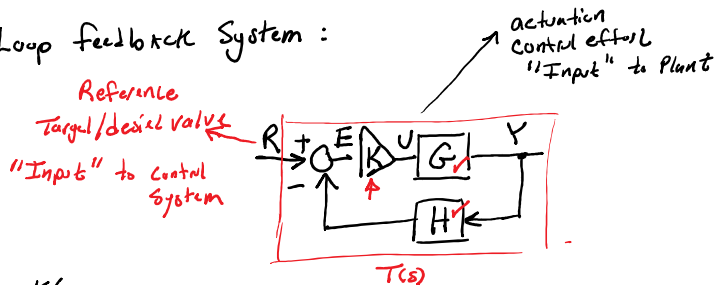


$\omega_n \sim$ Speed of Response
 $90\% \sim$ Stability

3) steady-state error



Closed Loop feedback System:



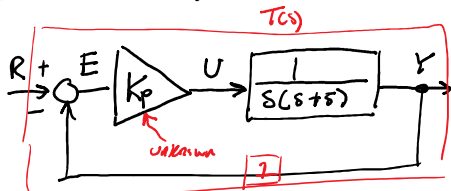
$$T(s) = \frac{Y}{R} = \frac{KG}{1+KGH}$$

Performance/stable? $\rightarrow$ Need Poles of System

$$1+KGH = 0 \quad \text{Characteristic Equation C.E.}$$

design K for desired Response.

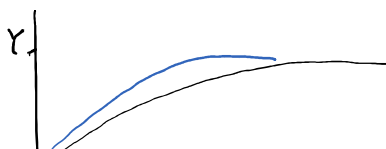
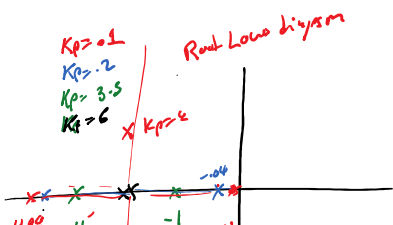
Example (Position Control)

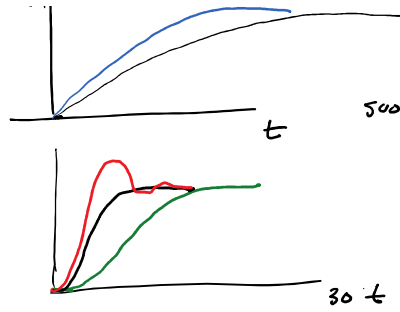
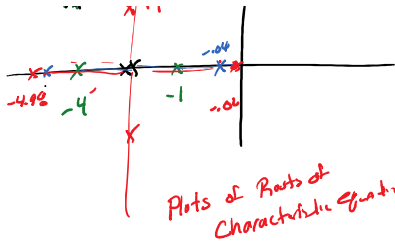


$$T(s) = \frac{Y}{R} = \frac{\frac{K_p}{s(s+5)}}{1 + \frac{K_p}{s(s+5)}} = \frac{K_p}{s(s+5) + K_p} = \frac{K_p}{s^2 + 5s + K_p}$$

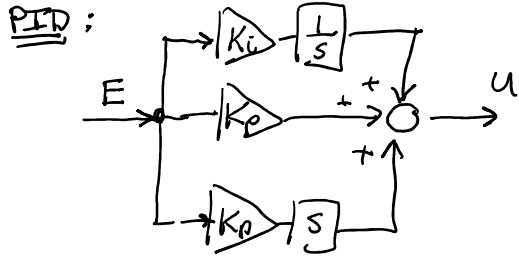
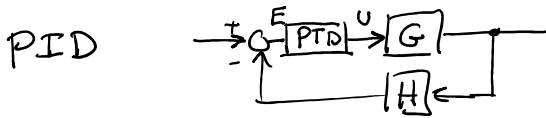
$$\text{C.E. } s^2 + 5s + K_p = 0 \quad \text{Poles of System}$$

Pick K_p to get desired Poles





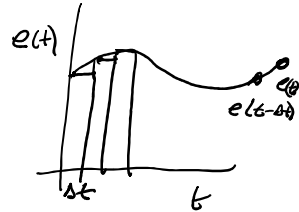
↑ Q4 handwritten



Equations:

$$u(t) = K_p e(t) + K_i \int e(t) dt + K_d \frac{de}{dt}$$

numerical Approximation
discrete equations



$$u = K_p e + K_i \sum e(t) \Delta t + K_d \frac{e(t) - e(t - \Delta t)}{\Delta t}$$

